

applications, active techniques might be too disturbing to the environment because of the additional heat that needs to be dumped. All such techniques add weight, bulk and some degree of complexity.

Operating electronics beyond the normal limits is thus an option worth considering. This special electronics will be able to withstand the extreme temperatures of space.

Extreme-temperature electronics

The term 'extreme-temperature electronics' (ETE) is used for electronics operating outside the traditional temperature range of -55°C to $+125^{\circ}\text{C}$. It covers both the very low temperatures—down to essentially absolute zero (0°K or -273°C)—and the high temperatures ($+125^{\circ}\text{C}$ and above).

In low-temperature electronics (LTE), operation of semiconductor-based devices and circuits has often been reported down to temperatures as low as a few degrees above absolute zero. These devices are based on silicon (Si), germanium (Ge), gallium arsenide (GaAs) and other semiconductor materials. Moreover, there is no reason to believe that operation might not extend all the way down to absolute zero.

In high-temperature electronics (HTE), laboratory operation of discrete semiconductor devices has been reported at temperatures as high as $+700^{\circ}\text{C}$ (for a diamond Schottky diode) and 650°C (for a silicon carbide (SiC) MOSFET). Integrated circuits (ICs) based on Si and GaAs have operated at 400 – 500°C . Si ICs have been reported to operate at $+300^{\circ}\text{C}$ for a thousand hours or longer. Covering both extremes, there are reports of the same transistor working at about -270°C to $+400^{\circ}\text{C}$ temperature range. Also, many passive components are usable to the lowest temperatures or up to several hundred degrees Celsius.

However, operation at extreme temperatures is not true for every

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semiconductor device or passive component; it depends on a number of material and design factors. Practical operation of devices and circuits is reasonably achievable to as low temperature as desired, provided materials and designs appropriate to the temperature are used. However, the various characteristics of the device might improve or degrade. In particular, below about 40°K (-230°C), Si devices often exhibit significant changes in characteristics.

High-temperature electronics presents more difficulty. The practical upper temperature limit is determined by many factors and the inherent temperature limit is often not reflected for semiconductor devices. The limit is frequently determined by the interconnections and packaging—both for active devices and passive components. As an indication of the practical upper limit, circuits have been offered commercially for operation at up to $+300^{\circ}\text{C}$.

Parts availability is a major obstacle to practical ETE. There are few components specified for either low- or high-temperature use. To construct ETE hardware, often the conventional-temperature-range components are selected and adapted. Custom fabrication is done if resources and time permit.

Space electronics over the years

In earlier days, space exploration devices were fragile and required constant human attention. Many of the space exploration missions were unsuccessful due to the failure of the incorporated electronic devices. As space race progressed, hardware had to be put into the orbit as swiftly as possible by making space electronics smaller, more compact and far more reliable.

Vacuum tubes were used as state-of-the-art electronics in the late 1950s. These were well understood and highly developed. Transistors were still pretty new then. Many of the transistors available did not work well at high frequencies as required in satellites. A high percentage of the computers used still depended on vacuum tubes.

Some of the earliest space probes used film cameras and transmitted the processed film in a manner similar to a fax machine. Another technique used in some later vehicles was 'slow-scan TV.' It involved a TV camera that took a single image and transmitted it, often very slowly.

It was realised further that better testing, redundancy and flexibility provided the keys to more reliable space electronics demanded by satellite operation.

Today, use of digital tape recorders and digital cameras to save data and pictures and transmit these to earth, or send commands to probes has been replaced with on-board electronic devices and computers. The first military satellite to have a microprocessor on board was the DSCS-III launched on October 30, 1982. Before that, ground commands were loaded into registers, and simpler circuits such as timers used to execute the commands. All parts must be able to survive launch, have radiation shielding, be able to radiate enough heat in a vacuum to function and have enough computing power to do the job.

Space exploration with space probes, space-shuttles, space crafts and satellites has proved effective with technological advances not only in the mechanical body parts, launch, guiding, and keeping the temperature of the space vehicle in check during the supersonic glide through the atmos-